

Official-source Electricity compilation 03

Mixed-source chapter worksheet • cross-paper reuse disclosed

Total: 34 marks • Suggested time: 68 minutes

Instructions and reuse disclosure

Curated compilation of official CBSE source questions, not a CBSE-issued chapter paper. Original source labels and exact source images are retained.

Answer every selected question. Only one selected source-slot alternative appears in this paper; do not answer or count excluded alternatives.

Questions recur across this 15-paper collection. Reuse is disclosed per question in the source appendix; these are not 15 disjoint official papers.

Sections use available source marks: A MCQ; B 2 marks; C 3 marks; D 5 marks or source-marked case questions. Case marks are not inflated to five.

Section A — MCQ

Compilation questions 1-4; 4 marks. Exact source labels remain attached to each question.

Section B — 2-mark questions

Compilation questions 5-8; 8 marks. Exact source labels remain attached to each question.

Section C — 3-mark questions

Compilation questions 9-12; 12 marks. Exact source labels remain attached to each question.

Section D — 5-mark / case-based questions

Compilation questions 13-14; 10 marks. Exact source labels remain attached to each question.

Section A — MCQ

1. [1 marks] Source Q13 • main

- 13.** A wire of length ' l ' is gradually stretched so that its length increases to $3l$.
If its original resistance is R , then its new resistance will be :

- | | |
|----------|-----------|
| (A) $3R$ | (B) $6R$ |
| (C) $9R$ | (D) $27R$ |

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

2. [1 marks] Source Q13 • 2022-23

13	A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire (a) decreases. (b) remains the same. (c) increases. (d) increases for some time and then decreases .	1
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3. [1 marks] Source Q14 • 2019-20

14 **Assertion:** A fuse wire is always connected in parallel with the mainline. 1

Reason: If a current larger than the specified value flows through the circuit, fuse wire melts.

For question numbers 13 and 14, two statements are given- one labeled *Assertion* (A) and the other labeled *Reason* (R). Select the correct answer to these questions from the codes (i), (ii), (iii) and (iv) as given below

- i) Both A and R are true and R is correct explanation of the assertion.
- ii) Both A and R are true but R is not the correct explanation of the assertion.
- iii) A is true but R is false.
- iv) A is false but R is true.

13 **Assertion:** Following are the structural isomers of butane. 1

Source/editorial note: Fuse series placement and melting from excess current are explicitly NCERT Electricity section 11.7.1 (printed p190, PDF p20). Included after independent audit overturned the batch reviewer's domestic-circuit exclusion. Shared assertion-reason choices occur on preceding PDF page 5.

4. [1 marks] Source Q15 • compartment

15. The resistance of a wire does ***not*** depend on its :

- (a) Length
- (b) Area of cross-section
- (c) Shape
- (d) Material

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

Section B — 2-mark questions

5. [2 marks] Source Q27 • main

The values of current (I) flowing through a given resistor of resistance (R), for the corresponding values of potential difference (V) across the resistor are as given below :

V (volts)	0.5	1.0	1.5	2.0	2.5	3.0	4.0	5.0
I (amperes)	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0

Plot a graph between current (I) and potential difference (V) and determine the resistance (R) of the resistor.

Source/editorial note: Occurrence mappings retained verbatim; only primary PDF/question evidence is accepted and counted. Other occurrences require their own evidence validation. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

6. [2 marks] Source Q34 VI A • 2025-26

For visually impaired students

- A. You are given four resistors each having resistance of R ohm. Find the maximum and minimum resistance that can be made with these four resistors.

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

7. [2 marks] Source Q34 A • 2026-27

34

Attempt either option A or B.

2

A.

- I. State Joule's Law of heating.
- II. What happens to the heat produced in a conductor if the resistance is doubled while keeping other variables constant?

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

8. [2 marks] Source Q25 VI B • 2024-25

OR

- B. You are given 2 fuse wires A and B with current ratings 2A and 5A respectively. Justify with reason which of the two would you use with a 1000W, 220V room heater?

Source/editorial note: Borderline household safety context retained by explicit scope instruction: electric-power/current calculation and fuse rating. Original MS wording "with is" retained. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Section C — 3-mark questions

9. [3 marks] Source Q36 • 2026 First Board



- 36.** The values of current I flowing in a given resistor for the corresponding values of potential difference V applied across the ends of resistor are given below in the table :

I (Amperes)	0.4	0.8	1.2	1.6	2.0
V (Volts)	1.2	2.4	3.6	4.8	6.0

Plot a graph between V and I and calculate the resistance of that resistor. 3

Source/editorial note: Visually confirmed complete replacement crop: question number, wording, all options/table values, mathematical symbols and printed marks. Whitespace normalized; mathematical glyphs represented in Unicode and stacked fractions as $\frac{1}{4}$ and $\frac{V}{4}$. Page/QR/footer debris removed from text; marks retained in marks field. No supplied diagram; table retained in text and evidence. Answers not reviewed. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

10. [3 marks] Source Q7 • 2017-18

	(a) Lights and fans:	(b) Appliances of 2kW or more power:	
7	Draw a circuit diagram of an electric circuit containing a cell, a key, an ammeter, a resistor of 4Ω in series with a combination of two resistors (8Ω each) in parallel and a voltmeter across parallel combination. Each of them dissipate maximum energy and can withstand a maximum power of 16W without melting. Find the maximum current that can flow through the three resistors.		3
8	In the electrolysis of water		2

Source/editorial note: Question asks the candidate to draw a circuit; no diagram is supplied in the SQP. Original wording "Each of them dissipate maximum energy" retained. In the connected circuit the 4Ω resistor limits the total current to 2 A; each 8Ω resistor then dissipates 8 W, not 16 W. The official answer follows this limiting-current interpretation. Final audit replaced clipped original crop; complete final "resistors." line visually confirmed.

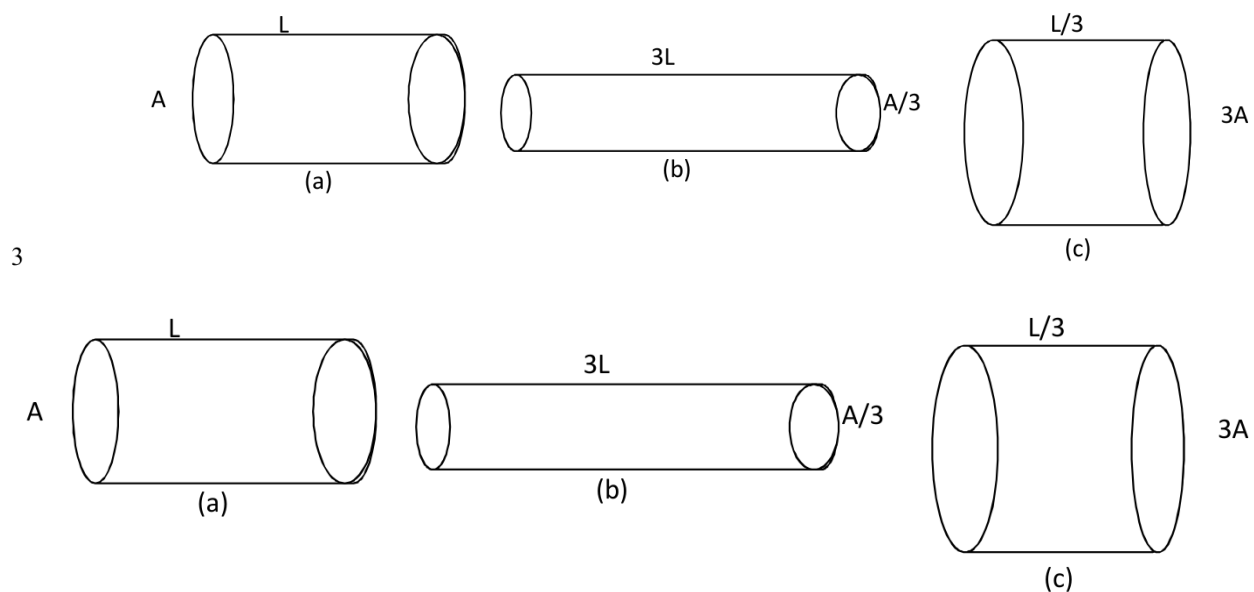
11. [3 marks] Source Q11 • 2021-22

11	a. It would cost a man Rs. 3.50 to buy 1.0 kW h of electrical energy from the Main Electricity Board. His generator has a maximum power of 2.0 kW. The generator produces energy at this maximum power for 3 hours. Calculate how much it would cost to buy the same amount of energy from the Main Electricity Board.(1 Mark) b. A student boils water in an electric kettle for 20 minutes. Using the same mains supply he wants to reduce the boiling time of water. To do so should he increase or decrease the length of the heating element? Justify your answer.(2 Marks)	3
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Source/editorial note: Both compulsory subparts retained in one three-mark record. MS answer spans pp.3-4.

12. [3 marks] Source Q13 • 2018-19

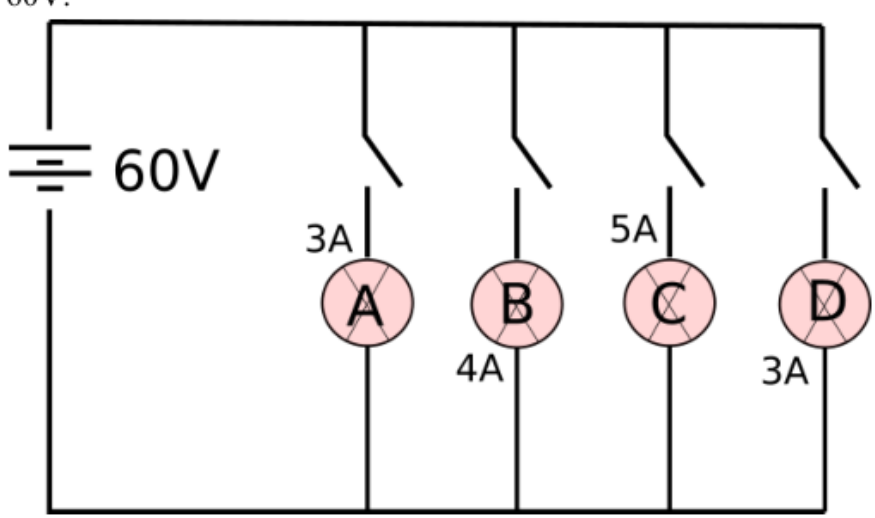
13. The figure below shows three cylindrical copper conductors along with their face areas and lengths. Compare the resistance and the resistivity of the three conductors. Justify your answer.

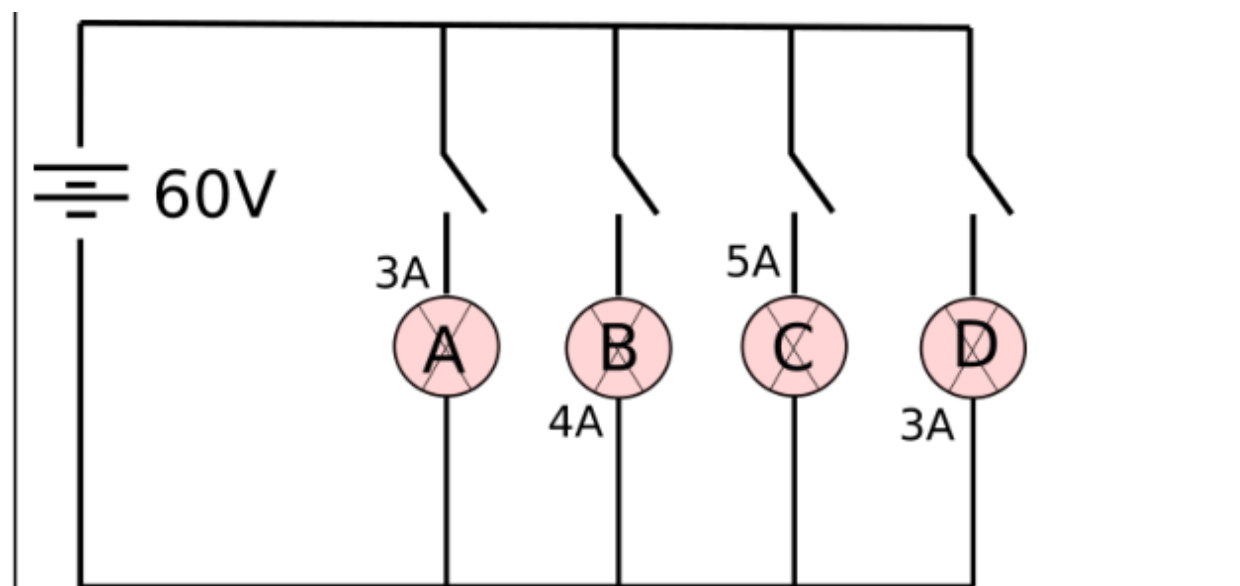


Source/editorial note: Diagram labels verified visually: (a) L and A ; (b) $3L$ and $A/3$; (c) $L/3$ and $3A$. The full question crop preserves a stray printed "3" below the figures as present in the source.

Section D — 5-mark / case-based questions

13. [5 marks] Source Q36 • 2020-21

36	In the given circuit, A, B, C and D are four lamps connected with a battery of 60V.  Analyse the circuit to answer the following questions. (i) What kind of combination are the lamps arranged in (series or parallel)? (ii) Explain with reference to your above answer, what are the advantages (any two) of this combination of lamps? (iii) Explain with proper calculations which lamp glows the brightest? (iv) Find out the total resistance of the circuit.	5
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Source/editorial note: All four subparts retained. The OR alternative on pp.14-15 concerns magnetic fields and the right-hand thumb rule and is excluded. Source MS p.8 prints " $= 60/15 \text{ A} = 4\text{A}$ "; the resistance unit is corrected explicitly to Ω , not silently attributed to the source.

14. [5 marks] Source Q36 • compartment

36. (a) (i) Write the relation between the electrical resistivity ' ρ ' and the resistance ' R ' of a cylindrical conductor of length ' l ' and area of cross-section ' A '. Use this relationship to determine the SI unit of ' ρ '.

(ii) Find the electrical resistivity of the material of a wire of length 120 cm and diameter 0.4 mm. The resistance of this wire is 3 Ω .

5

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — pyq-2025-89ec80e87c4adf229cc3e824

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 2 — sqp_2022-23_q13

(c) increases

13.	(c) increases	1
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Marking-scheme PDF page(s): [1]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

Question 3 — SQP-2019-20-Q14

iv) Assertion is false but reason is true.

14	iv) Assertion is false but reason is true.	1 mark
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Marking-scheme PDF page(s): [2]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

Question 4 — pyq-2023-dc74f0e84e53633b06097022

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 5 — pyq-2018-ac2b7b28762938a4a2df67a3

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 6 — SQP_2025-26_q34_VI_A

(i) Maximum Resistance:

To get the maximum resistance, connect all four resistors in series.

The total resistance $R_{\max}$ in series is the sum of the individual resistance:

$$R_{\max} = R + R + R + R = 4R$$

(ii) Minimum Resistance:

To get the minimum resistance, connect all four resistors in parallel.

The total resistance $R_{\min}$ in parallel is given by:

$$1/R_{\min} = 1/R + 1/R + 1/R + 1/R = 4/R \Rightarrow R_{\min} = R/4$$

For visually impaired students

A.

(i) Maximum Resistance:

- To get the maximum resistance, connect all four resistors in series.
- The total resistance R_{max} in series is the sum of the individual resistance:

$$R_{max} = R + R + R + R = 4R$$

(ii) Minimum Resistance:

- To get the minimum resistance, connect all four resistors in parallel.

- The total resistance R_{min} in parallel is given by:

$$\frac{1}{R_{min}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{4}{R}$$

$$R_{min} = \frac{R}{4}$$

Marking-scheme PDF page(s): [7, 8]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

Question 7 — SQP_2026-27_q34_A

A.

I. Joule's law of heating: The heat produced in a resistor is

(i) directly proportional to the square of current for a given resistance,

(ii) directly proportional to resistance for a given current, and

(iii) directly proportional to the time for which the current flows through the resistor.

$$H = I^2Rt,$$

where 'I' is current and 'R' is resistance. (1)

II. If resistance is doubled: For the same current I, heat H becomes

$$H = I^2(2R)t = 2I^2Rt = 2H,$$

i.e. heat doubles. (1)

34

Student to attempt either A or B.

2

A.

I. Joule's law of heating: The heat produced in a resistor is

(i) directly proportional to the square of current for a given resistance,

(ii) directly proportional to resistance for a given current, and

(iii) directly proportional to the time for which the current flows through the resistor.

$$H = I^2Rt,$$

where 'I' is current and 'R' is resistance. (1)

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$$H = I^2(2R)t = 2I^2Rt = 2H,$$

i.e. heat doubles. (1)

Marking-scheme PDF page(s): [7]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

Question 8 — SQP_2024-25_q25_VI_B

B. $P = V \times I$ (0.5)

$I = P/V = 1000/220 = 4.54 \text{ A}$ (0.5)

We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)

OR

B. $P = V \times I$ (0.5)

$I = \frac{P}{V} = \frac{1000}{220} = 4.54 \text{ A}$ (0.5)

We will be using fuse B with is rated as 5A. This is because it will be able to sustain the current (4.54 A) passing through it. Whereas fuse A will melt and break the circuit as the current exceeds its rating. (1)

Marking-scheme PDF page(s): [3]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

Question 9 — pyq-2026-c2c822cabb523e97bd204b9e

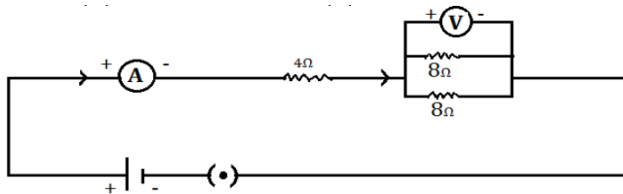
Official answer unavailable/withheld: No independently verified official answer supplied.

Original marking scheme (see limitation above): https://www.cbse.gov.in/cbsenew/marketing-scheme/2026/X/086_SCIENCE.zip

Question 10 — cbse_sqp_2017-18_electricity_q7

Circuit diagram as in MS page 1. Maximum current through 4Ω resistor = $\sqrt{(P/R)} = \sqrt{(16/4)} = 2 \text{ A}$.
Maximum current through each 8Ω resistor = $(1/2) \times 2 = 1 \text{ A}$.

7



1

Maximum current through 4Ω resistor $= \sqrt{\frac{P}{R}}$ 1

$$= \sqrt{\frac{16}{4}} = 2 \text{ A}$$

$\therefore$ Maximum current through each 8Ω resistor $= \frac{1}{2} \times 2 = 1 \text{ A}$ 1

Marking-scheme PDF page(s): [1, 2]

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_MS.pdf

Question 11 — sqp_2021-22_term2_q11

(a) $E = P \times T = 3 \times 2 = 6 \text{ kWh}$. Cost = $6 \times 3.50 = \text{Rs. } 21.0$. (b) To reduce the boiling time, the rate of heat production should be large. $P = V^2/R$; V is constant, so R should be decreased. Since R is directly proportional to length l , the length should be decreased.

11	a. $E = P \times T$ SO, $E = 3 \times 2 = 6 \text{ kWh}$ (1 mark) Cost of buying electricity from the main electricity board = $6 \times 3.50 =$ Rs. 21.0	3
	b. To reduce the boiling time using the same mains supply, the rate of heat production should be large. We know that $P = V^2/R$. Since V is constant, R should be decreased. Since R is directly proportional to l so length should be decreased. (2 marks)	

Marking-scheme PDF page(s): [3, 4]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-MS_Term2.pdf

Question 12 — cbse_sqp_2018-19_electricity_q13

$R_a = \rho L/A$; $R_b = \rho(3L)/(A/3) = 9\rho L/A = 9R_a$; $R_c = \rho(L/3)/(3A) = (1/9)\rho L/A = (1/9)R_a$. Hence $R_b > R_a > R_c$. $\rho_a = \rho_b = \rho_c$ because all three conductors are of the same material.

$$13. R_a = \rho \frac{l}{A} \quad 3$$

$$R_b = \rho \left(\frac{3L}{A/3} \right) = 9 \frac{\rho L}{A} = 9 R_a$$

$$R_c = \rho \frac{L/3}{3A} = \frac{1}{9} \frac{\rho L}{A} = \frac{1}{9} R_a$$

Hence $R_b > R_a > R_c$

$\rho_a = \rho_b = \rho_c$ because all the three conductors are of same material.

Marking-scheme PDF page(s): [4]

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

Question 13 — sqp_2020-21_q36_main

(i) The lamps are in parallel. (ii) A faulty lamp does not affect the working of the others; all lamps use the full potential of the battery. (iii) The lamp with the highest power glows brightest. All have the same voltage; C has the largest current. $P_C = 5 \times 60 = 300 \text{ W}$, the maximum. (iv) Total current = $3 + 4 + 5 + 3 = 15 \text{ A}$; $R = V/I = 60/15 = 4 \Omega$. The MS prints the final unit incorrectly as A.

36	The lamps are in parallel. Advantages: If one lamp is faulty, it will not affect the working of the other lamps. They will also be using the full potential of the battery as they are connected in parallel. The lamp with the highest power will glow the brightest. $P=VI$ In this case, all the bulbs have the same voltage. But lamp C has the highest current. Hence, for Lamp C $P=5 \times 60 \text{ Watt} = 300 \text{ W. (the maximum).}$ The total current in the circuit $= 3+4+5+3 \text{ A} = 15\text{A}$ The Voltage $= 60\text{V}$ $V=IR$ and hence $R = V/I$ $= 60/15 \text{ A} = 4\text{A}$	1 1 1 1 1
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Marking-scheme PDF page(s): [8]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-MS.pdf

Question 14 — pyq-2024-17a293a3889e84f2b1aa8c67

Official answer unavailable/withheld: No independently verified official answer supplied.

Per-question source appendix

Original CBSE questions/images remain attributed to their source. Compilation selection and numbering are editorial. The full Science source PDFs may include unrelated chapters; only selected Electricity crops form this worksheet.

1. pyq-2025-89ec80e87c4adf229cc3e824

CBSE — Class X Science — 2025 — main — 31/3/1 | PYQ | Year/session: 2025 / main | Paper code: 31/3/1 | Source question: 13 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 9

https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip

Also used in: official-04, official-13

2. sqp_2022-23_q13

CBSE Class X Science (086) Sample Question Paper 2022-23 | SQP | Year/session: 2022-23 / 2022-23 | Paper code: Not recorded | Source question: 13 | Original section: Section A | Selected marks: 1 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Also used in: official-05, official-06, official-09, official-13, official-15

3. SQP-2019-20-Q14

Science-086 Sample Question Paper 2019-20 Class-X | SQP | Year/session: 2019-20 / 2019-20 | Paper code: Not recorded | Source question: 14 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 5, 6

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Also used in: official-05, official-09, official-11, official-13

4. pyq-2023-dc74f0e84e53633b06097022

CBSE — Class X Science — 2023 — compartment | PYQ | Year/session: 2023 / compartment | Paper code: Not recorded | Source question: 15 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 11

<https://www.cbse.gov.in/cbsenew/question-paper/2023-COMPTT/X/Science.zip>

Also used in: official-02, official-06, official-09, official-11, official-14

5. pyq-2018-ac2b7b28762938a4a2df67a3

CBSE — Class X Science — 2018 — 31/1 — main | PYQ | Year/session: 2018 / main | Paper code: 31/1 | Source question: 27 | Original section: Section – B | Selected marks: 2 | PDF pages (one-based): 12

https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip

Also used in: official-09, official-13, official-15

6. SQP_2025-26_q34_VI_A

CBSE Class X Science (086) Sample Question Paper 2025-26 (English) | SQP | Year/session: 2025-26 / 2025-26 | Paper code: Not recorded | Source question: 34 VI A | Original section: Section – C | Selected marks: 2 | PDF pages (one-based): 12

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Also used in: official-06, official-11

7. SQP_2026-27_q34_A

CBSE Class X Science (086) Sample Question Paper 2026-27 (English) | SQP | Year/session: 2026-27 / 2026-27 | Paper code: Not recorded | Source question: 34 A | Original section: Section – C | Selected marks: 2 | PDF pages (one-based): 12

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Also used in: official-07

8. SQP_2024-25_q25_VI_B

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 25 VI B | Original section: Section-B | Selected marks: 2 | PDF pages (one-based): 7

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Also used in: official-08

9. pyq-2026-c2c822cabb523e97bd204b9e

CBSE Class X Science 2026 First Board 31-2-2 | PYQ | Year/session: 2026 / 2026 First Board | Paper code: 31/2/2 | Source question: 36 | Original section: Section C (Physics) | Selected marks: 3 | PDF pages (one-based): 29

<https://www.cbse.gov.in/cbsenew/question-paper/2026/X/Science.zip>

Also used in: official-15

10. cbse_sqp_2017-18_electricity_q7

Sample Question Paper 2017-18 — Science — Class - X | SQP | Year/session: 2017-18 / 2017-18 | Paper code: Not recorded | Source question: 7 | Original section: SECTION - A | Selected marks: 3 | PDF pages (one-based): 1

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf

Also used in: official-08

11. sqp_2021-22_term2_q11

CBSE Class X Science (086) Sample Question Paper 2021-22 — Term 2 | SQP | Year/session: 2021-22 / 2021-22 | Paper code: Not recorded | Source question: 11 | Original section: Section B | Selected marks: 3 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-SQP_Term2.pdf

Also used in: official-11

12. cbse_sqp_2018-19_electricity_q13

Class-X Science-086 SAMPLE QUESTION PAPER-19 | SQP | Year/session: 2018-19 / 2018-19 | Paper code: Not recorded | Source question: 13 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 3

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

Also used in: official-08, official-14

13. sqp_2020-21_q36_main

CBSE Class X Science (086) Sample Question Paper 2020-21 | SQP | Year/session: 2020-21 / 2020-21 | Paper code: Not recorded | Source question: 36 | Original section: Section D | Selected marks: 5 | PDF pages (one-based): 14

https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Also used in: official-05, official-08, official-12

14. pyq-2024-17a293a3889e84f2b1aa8c67

CBSE — Class X Science — 2024 — compartment — 31/S/1 | PYQ | Year/session: 2024 / compartment | Paper code: 31/S/1 | Source question: 36 | Original section: SECTION D | Selected marks: 5 | PDF pages (one-based): 23

<https://www.cbse.gov.in/cbsenew/question-paper/2024-COMPTT/X/SCIENCE.zip>

Also used in: official-04, official-06, official-13, official-15